25900787 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCCSC100, MG1DSCECC101 - ART OF COMPUTING AND PROBLEM SOLVING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer **All** Questions

Each Question carries **1** mark

1. If a variable is used before it is initialized, what type of error is likely to occur? [U] / [CO1]
2. Identify the problem-solving approach that involves combining small, well-defined modules to create a larger system. [K] / [CO1]
3. A program is written using binary codes (0s and 1s). Identify the type of language this refers to. [K] / [CO1]
4. How do you format a floating-point number to display two decimal places in C? [U] / [CO2]
5. Identify the operator in C that is used as a short form of the if-else statement. [K] / [CO2]
6. A programmer declares a variable with the identifier '1stValue' and initializes it to 10. What will happen when compiling this code? [U] / [CO2]
7. When does an ‘else’ block will be executed in an ‘if-else’ statement? [U] / [CO3]
8. When does a 'do-while' loop terminate? [U] / [CO3]

9. How do you skip to the next iteration of a loop? [U] / [CO3]

10. Explain what happens if the condition in a 'while' loop is always true. [U] / [CO3]

[10x1 = 10]

Part B

Short Answer Questions

Answer any **4** Questions

Each Question carries **5** marks

11. Explain the relationship between bits and bytes and their significance in computing. [K] / [CO1]

12. Discuss the purpose of testing and verification in the software development process. [U] / [CO1]

13. Discuss the fundamental data types in C with examples. [K] / [CO2]

14. How to call a function in C? Explain with an example. [K] / [CO3]

15. Write a C program to reverse a string. [A] / [CO3]

16. Write a program to find sum of elements of a one dimensional array. [A] / [CO3]

[4x5 = 20]

Part C

Essay Questions

Answer any **2** Questions

Each Question carries **10** marks

17. Describe the main functions of compilers, interpreters, and assemblers in programming and their roles in code translation. [U] / [CO1]

18. Discuss the use of varibles and constants in programming. Explain different types of constants with examples. [U] / [CO2]

19. Explain the syntax and functioning of a 'for' loop in C. Implement a 'for' loop for the following scenario: [A] / [CO3]
    A mathematics teacher wants to identify Armstrong numbers between 100 and 500 (inclusive).
    1. Write a C program using a 'for' loop to iterate through numbers from 100 to 500.
    2. Within the loop, check if each number is an Armstrong number.
    3. If a number is an Armstrong number, print it.

[2x10 = 20]